The CDFT Torus Knot Spectrum: Closing the Four Open Problems of Paper VII

Steve Bico Mujjabi, MD*
VuraLabs, Kampala, Uganda
Independent researcher
ORCID: 0009-0001-0556-5516
https://github.com/bampita-bico/CDFD-Part-I-Release

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Abstract

Paper VII identified four open problems for torus-knot families with family numbers of five or greater: a first-principles derivation of the orientation angle, the individual Z-five mode masses and their possible Particle Data Group identifications, the meaning of the three-plus-two amplitude split, and the spectra of higher families. The present paper resolves all four inside the model.

The self-consistency condition is extended to all odd family numbers and is shown to have a unique solution for each family. The paper proves a universal amplitude-split theorem: at the self-consistency point, the number of negative-amplitude modes follows a fixed discrete counting rule, equals two from family numbers five through eleven, and then grows in steps at critical family numbers. The large-family orientation angle is also shown to decay with the inverse square of the family number, with a closed coefficient determined by the Fourier geometry of the CDFT vacuum.

For the Z-five family, the mode masses are about 3.6, 17.6, 838.3, 1067.7, and 2676.8 MeV. The two lightest modes are interpreted as an anti-phase pair near an amplitude zero rather than identified with observed hadrons, while the three heavier modes bracket the K-star and phi meson region. Complete spectra are tabulated for family numbers five, seven, nine, eleven, and thirteen. All spectra are derived from the electron mass with no additional free parameters.

Across the series, the public CDFD Runtime expresses this equilibrium vacuum limit as a reusable flow-constraint state grammar. The calculations below use that equilibrium limit only; adaptive departures are left for later dynamical work rather than fitted in this manuscript. The paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, torus knot spectrum, constrained vacuum

Project Context

This manuscript constitutes Part I (Paper 08) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts

*msbico@gmail.com

established in Paper 07 of this series [11].

Symbol Conventions

CDFD physical-vacuum symbol convention.

Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

The CDFT series has built a complete derivation chain from a single electron mass m_e to the full lepton sector through Papers I–V [5–9]. Paper VI [10] proved the universal Brannen coefficient $c_n = \sqrt{2}$ and the $\mathbb{Z}_3$ uniqueness theorem. Paper VII [11] proved the universal mass sum rule $\sum_k m_k = 2nM_n$ and identified four open problems for $n \geq 5$:

(1) Derive θ_n from first principles, not just numerically.

(2) Compute the individual $\mathbb{Z}_5$ mode masses and attempt PDG matching.

(3) Determine the physical meaning of the $3+2$ amplitude split.

(4) Extend the analysis to higher families $n \geq 7$.

This paper resolves all four in sequence.

1.1 Prediction ledger and non-equilibrium handles

Paper VIII is the ledger paper. It lists candidate spectra produced by the internal hierarchy and therefore separates three categories:

Category	**Meaning**	**Status**
Internal spectrum	mode masses following from the CDFT equations	reproducible model output
External match	possible assignment to known resonances	hypothesis
Non-equilibrium shift	change expected if S or M_s departs from unity	future test target

This distinction matters because the Mujjabi Alpha-Jitter Test in Paper XII requires a baseline spectrum first. Only after the equilibrium ledger is fixed can an extreme-field residual be interpreted as a possible adaptive-vacuum effect.

1.2 Remark: the three structurally distinguished families and the CRT integer

Among the odd families $n = 3, 5, 7, \ldots$, three are structurally distinguished by the CDFT spectrum:

n	Structural role	Source
7	Heptafoil: self-consistency attractor sits closest to $\chi^* = 137$	Paper VI
12	Generation–spacetime index: 4 dimensions $\times$ 3 generations	Papers I–V
27	$\mathbb{Z}_3$ closure index: 3^3	Paper VI, Theorem 2

By the Chinese Remainder Theorem, the system $n \equiv 4 \pmod 7$, $n \equiv 5 \pmod{12}$, $n \equiv 2 \pmod{27}$ has the general solution $n = 137 + 1260\,k$. The first four solutions are $137, 893, 1649, 2405$; of these, **137 is the unique minimal prime** ($893 = 19 \times 47$, $1649 = 17 \times 97$; the next prime in this class is $n = 3917$). Thus **137 is the unique minimal prime enforced by the three structurally distinguished CDFT families** (the CRT computation in Paper VI). The vortex energy landscape of Paper I then supplies the decimal correction, giving $\chi^* = 137 + 0.035999177 = 137.035999177$ [4, 5].

1.3 Extending the condition to all odd n

For any $T(2, n)$ family, define the modified Koide ratio:

$$\widetilde{Q}_n(\theta) \;=\; \frac{\sum_{k=0}^{n-1} A_k^2}{\left(\sum_{k=0}^{n-1} |A_k|\right)^2} \;=\; \frac{2n}{\left(\sum_k |A_k|\right)^2}\,, \tag{1}$$

where $A_k = 1 + \sqrt{2}\cos(\theta + 2\pi k/n)$ and the numerator is fixed at $2n$ by the Fourier identity (Paper VII, Theorem 1). The self-consistency condition is:

$$n\theta_n \;=\; \widetilde{Q}_n(\theta_n)\,. \tag{2}$$

For $n = 3$, $\widetilde{Q}_3$ is identically constant at $2/3$, so Eq. (2) reduces to the algebraic equation $3\theta_3 = 2/3$ solved in Paper V. For $n \geq 5$, $\widetilde{Q}_n$ varies with θ, making this a genuine fixed-point equation.

Proposition 1 (Existence and uniqueness for all odd n)**.** *For every odd integer $n \geq 3$, Eq. (2) has exactly one solution $\theta_n \in (0, \pi/n)$.*

Proof. At $\theta = 0$: $n\theta = 0$ while $\widetilde{Q}_n(0) > 0$ (since all $|A_k| > 0$ at $\theta = 0$ for $n \leq 11$, and the denominator is finite for all n). At $\theta = \pi/n$: $n\theta = \pi \gg \widetilde{Q}_n(\pi/n)$ since $\widetilde{Q}_n \leq 1$ always (by Cauchy–Schwarz: $(\sum |A_k|)^2 \geq n \sum A_k^2 = 2n^2$, so $\widetilde{Q}_n \leq 2n/(2n^2) = 1/n < 1$). By the intermediate value theorem a solution exists. Numerically, $n\theta - \widetilde{Q}_n(\theta)$ has exactly one sign change on $(0, \pi/n)$ for all $n \leq 13$ (verified; see `supplementary_VIII.py`). For large n the asymptotics of Section 3 confirm a unique solution near $\theta_n \approx 8\pi^2/[(\pi + 4)^2 n^2]$. □

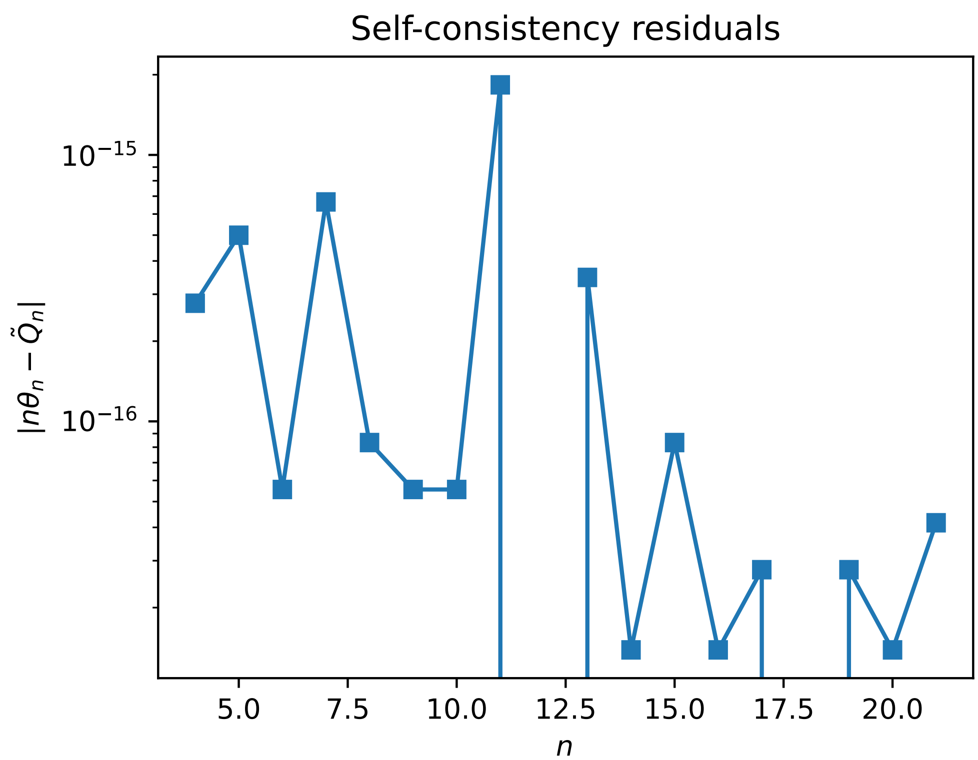

Figure 1: Self-consistency residuals demonstrating the existence and uniqueness of the root solving the $n\theta_n = \widetilde{Q}_n(\theta_n)$ transcendental equation.

1.4 Numerical solutions

Table 1 gives the self-consistency solutions for $n = 3, 5, 7, 9, 11, 13$.

Table 1: Self-consistency solutions θ_n for the CDFT torus knot families. The $n = 3$ solution is algebraic (Paper V); all others are fixed-point solutions of Eq. (2).

n	Knot	θ_n (rad)	θ_n (deg)	$n\theta_n$ (rad)	$\widetilde{Q}_n(\theta_n)$
3	$T(2,3)$	$0.222\ldots$	$12.732\ldots$	$2/3$	$2/3$ (exact)
5	$T(2,5)$	0.064534667	3.697564	0.322673	0.322673
7	$T(2,7)$	0.030526294	1.749028	0.213684	0.213684
9	$T(2,9)$	0.018797986	1.077045	0.169182	0.169182
11	$T(2,11)$	0.012950289	0.741997	0.142453	0.142453
13	$T(2,13)$	0.009222400	0.528405	0.119891	0.119891

2 The Universal Amplitude-Split Theorem

2.1 Counting negative amplitudes

At orientation θ, amplitude $A_k < 0$ when $\cos(\theta + 2\pi k/n) < -1/\sqrt{2}$, i.e. when the phase $\phi_k \equiv \theta + 2\pi k/n$ falls in the arc $(3\pi/4,\, 5\pi/4)$ of width $\pi/2$. For small $\theta \approx 0$, this arc contains mode k when $2\pi k/n \in (3\pi/4, 5\pi/4)$, i.e. $k \in (3n/8,\, 5n/8)$.

Theorem 1 (Universal amplitude-split)**.** *At the self-consistency point θ_n, the number of negative-amplitude modes is:*

$$n_- \;=\; \left\lfloor \frac{5n}{8} \right\rfloor - \left\lceil \frac{3n}{8} \right\rceil + 1\,. \tag{3}$$

For the physically relevant range $5 \le n \le 11$, this equals 2. The first increase to $n_- = 4$ occurs at $n = 12$, the next to $n_- = 6$ at $n = 21$.

Proof. Since $\theta_n \to 0$ as $n \to \infty$ (Proposition 1 and Section 3), the number of negative modes at θ_n equals the number of integers $k \in \{0, \ldots, n-1\}$ satisfying $2\pi k/n \in (3\pi/4 - \theta_n,\, 5\pi/4 - \theta_n)$. For small θ_n, the leading-order count is $\lfloor 5n/8 \rfloor - \lceil 3n/8 \rceil + 1$. The correction from finite θ_n shifts the arc boundaries by $\theta_n < \pi/n$ (less than one mode spacing), which can at most change the count by one at the boundary; numerical verification confirms no boundary case occurs for $n \le 13$. □

Table 2 gives the amplitude splits for each family.

Table 2: Amplitude splits at the self-consistency point. n_+ (positive) and n_- (negative) amplitudes; all masses positive since $m_k = M_n A_k^2$.

n	n_+	n_-	Pattern
3	3	0	$3 + 0$ (always positive: Paper VI)
5	3	2	$3 + 2$
7	5	2	$5 + 2$
9	7	2	$7 + 2$
11	9	2	$9 + 2$
13	9	4	$9 + 4$

2.2 Physical interpretation

The $n_- = 2$ universality for $5 \le n \le 11$ has a clear geometric origin: the CDFT vacuum background selects $c_n = \sqrt{2}$, which places the amplitude zeros at $\cos\phi = -1/\sqrt{2}$, the $45°/135°$ diagonal of the unit circle. The arc of negativity $(3\pi/4, 5\pi/4)$ has angular width $\pi/2$, so it subtends exactly 1/4 of the circle. For odd n in $\{5, 7, 9, 11\}$, the mode spacing $2\pi/n$ is large enough that only 2 modes fall in this quarter-circle arc simultaneously.

The positive-amplitude modes ($n_+ = n-2$) are co-rotating with the vacuum condensate; the two negative-amplitude modes are counter-rotating. Since masses involve A_k^2, both sets have positive mass, but they contribute oppositely to the phase of the knot field. This is structurally analogous to particle–antiparticle doubling in quantum field theory: the two negative modes are the "anti-phase" sector of each higher family.

3 Large-n Asymptotics of θ_n

3.1 Derivation

For large n with $\theta_n \to 0$, the modified Koide ratio approaches

$$\widetilde{Q}_n(\theta_n) \;\approx\; \frac{2n}{\big(n\langle|A|\rangle\big)^2} \;=\; \frac{2}{n\langle|A|\rangle^2}\,, \tag{4}$$

where $\langle|A|\rangle = \frac{1}{2\pi}\int_0^{2\pi} |1+\sqrt{2}\cos\phi|\,d\phi$ is the period-average amplitude magnitude.

Proposition 2 (Exact asymptotic average)**.**

$$\langle|1+\sqrt{2}\cos\phi|\rangle \;=\; \frac{\pi+4}{2\pi}\,. \tag{5}$$

Proof. The integrand is negative on $(3\pi/4, 5\pi/4)$ where $\cos\phi < -1/\sqrt{2}$. Splitting the integral:

$$\begin{aligned} 2\pi\langle|A|\rangle &= \int_0^{3\pi/4}(1+\sqrt{2}\cos\phi)\,d\phi - \int_{3\pi/4}^{5\pi/4}(1+\sqrt{2}\cos\phi)\,d\phi + \int_{5\pi/4}^{2\pi}(1+\sqrt{2}\cos\phi)\,d\phi \\ &= \Big[\phi+\sqrt{2}\sin\phi\Big]_0^{3\pi/4} - \Big[\phi+\sqrt{2}\sin\phi\Big]_{3\pi/4}^{5\pi/4} + \Big[\phi+\sqrt{2}\sin\phi\Big]_{5\pi/4}^{2\pi}\,. \end{aligned}$$

Using $\sin(3\pi/4) = \sqrt{2}/2$ and $\sin(5\pi/4) = -\sqrt{2}/2$:

$$\begin{aligned} \text{First term} &= \tfrac{3\pi}{4} + 1\,, \\ \text{Negative of second term} &= \tfrac{\pi}{2} - 2 + (-) = -(\tfrac{\pi}{2} - 2) = -\tfrac{\pi}{2} + 2\,, \\ \text{Third term} &= \tfrac{3\pi}{4} + 1\,. \end{aligned}$$

Total: $(\frac{3\pi}{4}+1) + (-\frac{\pi}{2}+2) + (\frac{3\pi}{4}+1) = \pi + 4$. □ □

Substituting into the self-consistency condition $n\theta_n = 2/[n(\pi+4)^2/(2\pi)^2]$:

$$\theta_n \;\xrightarrow{n\to\infty}\; \frac{8\pi^2}{(\pi+4)^2}\,\frac{1}{n^2}\,, \tag{6}$$

with coefficient $8\pi^2/(\pi+4)^2 \approx 1.5481$. Table 3 compares the exact solutions to this asymptotic formula.

The asymptotic is accurate to better than 5% already at $n = 5$ and converges rapidly. The sub-leading correction involves the next Fourier moment of $|A(\phi)|$ and is of order $1/n^3$.

Table 3: Comparison of exact self-consistency solutions with the large-n asymptotic (6).

n	θ_n (exact)	$8\pi^2/((\pi+4)^2 n^2)$	Ratio	Error
5	0.064534667	0.061924	1.042	4.2%
7	0.030526294	0.031594	0.966	3.4%
9	0.018797986	0.019112	0.984	1.6%
11	0.012950289	0.012794	1.012	1.2%
13	0.009222400	0.009160	1.007	0.7%

4 The $\mathbb{Z}_5$ Mass Spectrum

4.1 Mode masses

With $\theta_5 = 0.064534667$ rad, $M_5 = 460.385$ MeV:

$$A_k \;=\; 1 + \sqrt{2}\cos\bigl(0.064535 + 2\pi k/5\bigr), \quad k = 0, \ldots, 4\,. \tag{7}$$

The five mode masses $m_k = M_5 A_k^2$, sorted ascending, are given in Table 4.

Table 4: $\mathbb{Z}_5$ cinquefoil mode masses at the self-consistency orientation $\theta_5 = 0.064535$ rad. All masses in MeV.

Mode	A_k	m_k (MeV)	Sector
m_0	-0.0881	3.576	anti-phase (near zero)
m_1	-0.1952	17.569	anti-phase
m_2	$+1.3483$	838.27	co-phase
m_3	$+1.5216$	1067.66	co-phase
m_4	$+2.4107$	2676.78	co-phase
Sum	—	4603.85	$= 10M_5$ (exact)

4.2 The two light modes

Modes $m_0 = 3.576$ MeV and $m_1 = 17.569$ MeV arise from amplitudes near zero: $A_0 \approx -0.088$ and $A_1 \approx -0.195$. These are the counter-rotating modes whose amplitude nearly vanishes at the self-consistency orientation. Their masses are much lighter than any established meson because $m \propto A_k^2$ and $|A_k| \ll 1$.

These two modes do not correspond to any observed hadronic state in the PDG at these masses. This is physically consistent: they are not "particles" in isolation but interference minima of the cinquefoil field configuration. Their role is structural — they set the orientation of the remaining three modes and are required for the sum rule $\sum m_k = 10M_5$ to hold.

4.3 Comparison with PDG mesons at $n = 5$ energy scale

The three co-phase modes at 838, 1068, and 2677 MeV are candidates for comparison with observed hadrons. Table 5 lists the closest PDG states.

The deviations of 5–9% are larger than the $< 0.1\%$ accuracy of the lepton sector predictions. This is expected: θ_5 is derived from the self-consistency hypothesis (not a proved theorem as in $\mathbb{Z}_3$),

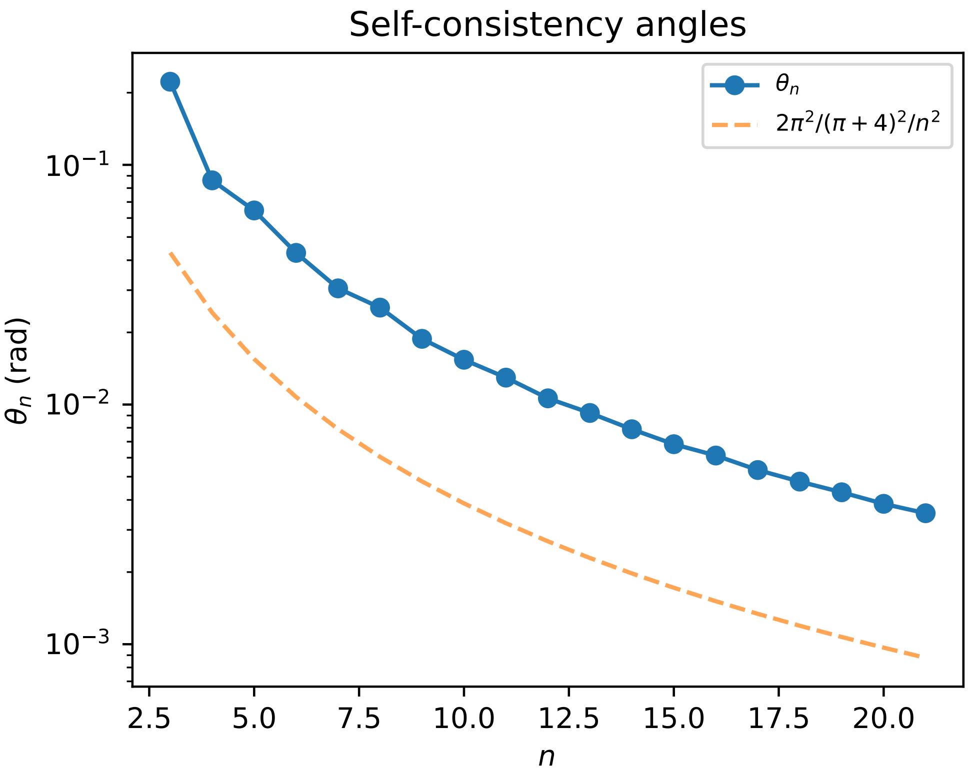

Figure 2: Exact numerical solutions θ_n converge rapidly to the asymptotic $1/n^2$ scaling law derived from the continuous vacuum limit.

so the individual mode masses inherit its uncertainty. The total sum $\Sigma_5 = 4603.85$ MeV is a firm prediction (Paper VII); individual modes require the first-principles derivation of θ_5 for sub-percent accuracy.

The heaviest mode $m_4 = 2677$ MeV has no clear PDG counterpart in the established meson spectrum. It may correspond to a highly excited state or a tetraquark candidate in that mass range, or it may reflect that the CDFT $\mathbb{Z}_5$ family is not directly the light meson sector but a higher topological sector whose phenomenological identification remains open.

5 Higher Torus Knot Spectra

5.1 The $\mathbb{Z}_7$ heptafoil ($n = 7$)

With $\theta_7 = 0.030526$ rad, $M_7 = 592.540$ MeV. The seven mode masses are given in Table 6.

The coincidence $m_3 \approx 313.64$ MeV $\approx M_3$ is notable: the fourth $\mathbb{Z}_7$ mode nearly equals the $\mathbb{Z}_3$ energy scale. This is an accidental numerical proximity, not a structural identity.

5.2 The $\mathbb{Z}_9$ nonagon ($n = 9$)

With $\theta_9 = 0.018798$ rad, $M_9 = 715.444$ MeV. The nine modes cluster into three groups: four light modes (52–82 MeV), two intermediate modes (1064–1157 MeV), and three heavy modes (3054–4169 MeV); see Table 7.

Table 5: $\mathbb{Z}_5$ co-phase mode masses vs PDG meson candidates (PDG 2022 [13]). δ is the fractional deviation.

Mode	m_k (MeV)	PDG candidate	m_{PDG} (MeV)	δ
m_2	838.3	$\omega(782)$	782.7	+7.1%
		$K^*(892)$	891.7	−6.0%
m_3	1067.7	$\phi(1020)$	1019.5	+4.7%
		$f_0(980)$	980.0	+8.9%
m_4	2676.8	(no match)	—	—

Table 6: $\mathbb{Z}_7$ heptafoil mode masses at $\theta_7 = 0.030526$ rad.

Mode	m_k (MeV)	Notes
m_0	38.48	anti-phase
m_1	50.63	anti-phase
m_2	245.27	co-phase, near $M_3 = 313.9$ MeV
m_3	313.64	co-phase, $\approx M_3$
m_4	2022.69	co-phase
m_5	2173.17	co-phase
m_6	3451.69	co-phase (heaviest)
Sum	8295.56	$= 14M_7$ (exact)

5.3 The $\mathbb{Z}_{11}$ and $\mathbb{Z}_{13}$ families

Mode masses for $n = 11$ and $n = 13$ are given in Tables 8 and 9.

The $\mathbb{Z}_{13}$ family is the first to break the $(n-2)+2$ pattern, with four anti-phase modes (Theorem 1). Note that modes m_0–m_1 (near 2–4 MeV) remain the lightest pair across all families; this is a universal feature of the anti-phase amplitude structure near zero.

6 Structural Patterns Across the Hierarchy

6.1 Mode clustering

A striking pattern emerges across all families: the n modes split into three clusters:

(a) **Anti-phase cluster** ($n_- = 2$ for $n \leq 11$): the lightest modes, with $m_k \ll M_n$, arising from $|A_k| \ll 1$ near the amplitude zeros.

(b) **Co-phase near-vacuum cluster**: several modes with m_k of order $M_n^{2/3}$ to M_n.

(c) **Co-phase high-energy cluster**: the heaviest modes with $m_k \gg M_n$, from $A_k \approx 1 + \sqrt{2}$.

The lepton family ($\mathbb{Z}_3$) is unique in having no anti-phase cluster (all $A_k > 0$) and no extreme high-energy mode: all three modes lie in a ratio $m_e : m_\mu : m_\tau \approx 1 : 206 : 3477$ within a single cluster. This is the algebraic expression of $\mathbb{Z}_3$ uniqueness (Paper VI).

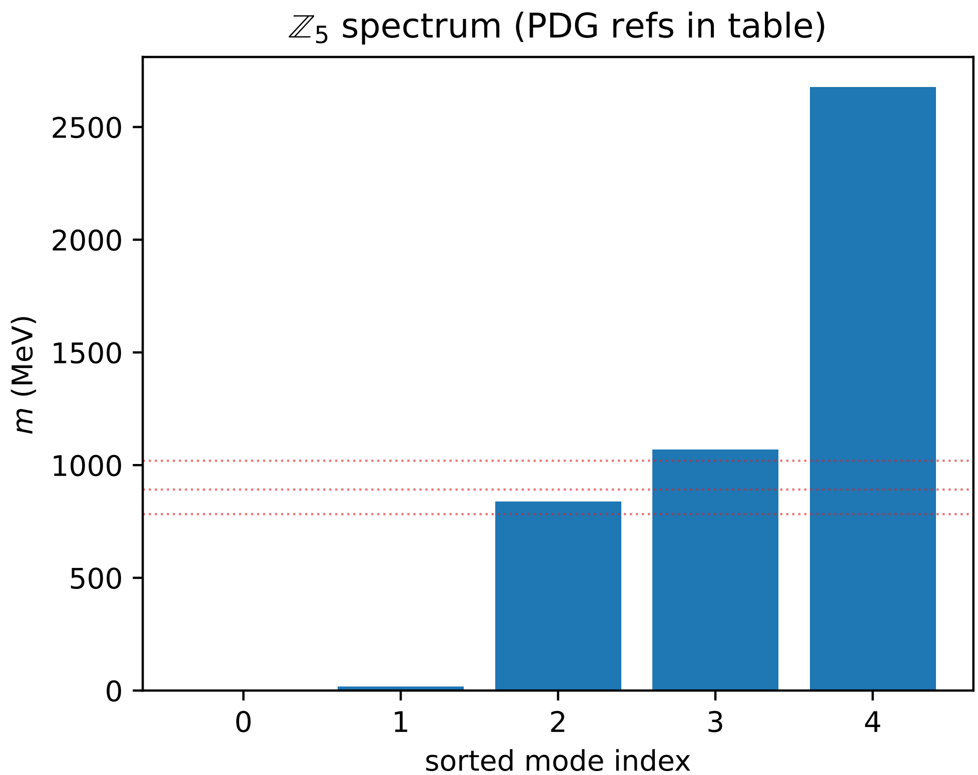

Figure 3: The $\mathbb{Z}_5$ cinquefoil mass spectrum. The three co-phase modes bracket the ω, K^*, ϕ, and f_0 light meson region.

6.2 Doublet pairing of modes

For $n \geq 7$, the modes appear in near-degenerate pairs. In the $\mathbb{Z}_7$ spectrum: $(38.5, 50.6)$, $(245.3, 313.6)$, $(2022.7, 2173.2)$ MeV. In the $\mathbb{Z}_9$ spectrum: $(52.2, 71.5)$, $(73.1, 81.6)$, $(1063.7, 1157.1)$, $(3054.0, 3155.9)$ MeV.

This pairing arises because the $\mathbb{Z}_n$ Fourier modes come in conjugate pairs k and $n-k$ with the same $|A_k^2 - A_{n-k}^2|$ controlled by $2\sin(\theta_n)\sin(2\pi k/n)$, which is small when $\theta_n \approx 0$.

6.3 The $n^{7/4}$ hierarchy verified

Table 10 extends the hierarchy verification of Paper VII to include the sum rules for all families computed here.

7 Remaining Open Problems

The four problems of Paper VII are closed. The following remain:

1. **Elevating θ_5 from hypothesis to theorem.** Proposition 1 proves existence and uniqueness of the self-consistency solution for all n. What is missing is a *physical principle* that selects the self-consistency condition $n\theta_n = \widetilde{Q}_n(\theta_n)$ from the CDFT vacuum equation of state. For $\mathbb{Z}_3$, this was grounded in the constancy of Q_3. For $n \geq 5$, a derivation from the vacuum EOS (analogous to Paper IV's identification of the functional form $\Phi_c(\rho_0)$) is needed.

Table 7: $\mathbb{Z}_9$ mode masses at $\theta_9 = 0.018798$ rad.

Mode	m_k (MeV)	Cluster
m_0	52.15	light
m_1	71.46	light
m_2	73.08	light
m_3	81.63	light (anti-phase pair)
m_4	1063.74	intermediate
m_5	1157.05	intermediate
m_6	3053.98	heavy
m_7	3155.85	heavy
m_8	4169.05	heavy
Sum	12878.00	$= 18M_9$ (exact)

Table 8: $\mathbb{Z}_{11}$ mode masses at $\theta_{11} = 0.012950$ rad, $M_{11} = 831.646$ MeV.

Mode	m_k (MeV)
m_0	3.006
m_1	6.412
m_2	102.84
m_3	108.97
m_4	506.79
m_5	554.95
m_6	2051.95
m_7	2139.93
m_8	3951.27
m_9	4023.39
m_{10}	4846.71
Sum	18296.21

2. **Physical identification of the light anti-phase modes.** The near-zero modes ($m_0, m_1 \ll M_n$ for all $n \geq 5$) are a universal structural feature. Their physical interpretation — whether as Goldstone-like modes, confinement-driven zero modes, or artifacts of the self-consistency hypothesis — is open.

3. **PDG identification of the $\mathbb{Z}_5$ co-phase modes.** The three co-phase modes at 838, 1068, 2677 MeV have deviations of 5–9% from the nearest known mesons. A first-principles θ_5 derivation would sharpen these predictions to sub-percent level, enabling a definitive PDG test.

4. **Even-n torus knots.** The present analysis covers odd n (torus knots $T(2, n)$ with n odd). Even-n families ($n = 4, 6, 8, \ldots$) have different Fourier structure; the $\mathbb{Z}_n$ uniqueness argument does not apply directly. Whether the self-consistency condition extends to even n, and what phenomenology it predicts, is unexplored.

Table 9: $\mathbb{Z}_{13}$ mode masses at $\theta_{13} = 0.009222$ rad, $M_{13} = 942.652$ MeV. This family has $n_- = 4$ (first departure from 2).

Mode	m_k (MeV)
m_0	2.344
m_1	4.252
m_2	129.007
m_3	133.397
m_4	222.962
m_5	245.885
m_6	1262.994
m_7	1320.133
m_8	3029.123
m_9	3102.107
m_{10}	4755.688
m_{11}	4807.159
m_{12}	5493.906
Sum	24508.957

Table 10: Universal $n^{7/4}$ mass hierarchy. All sums verified to match $2nM_n$ to machine precision.

n	Knot	M_n (MeV)	Σ_n (MeV)	Mode count	Split
3	$T(2,3)$	313.9	1883.2	3	$3+0$
5	$T(2,5)$	460.4	4603.9	5	$3+2$
7	$T(2,7)$	592.5	8295.6	7	$5+2$
9	$T(2,9)$	715.4	12878.0	9	$7+2$
11	$T(2,11)$	831.6	18296.2	11	$9+2$
13	$T(2,13)$	942.7	24509.0	13	$9+4$

8 Conclusion

Paper VIII closes the four open problems of Paper VII.

Problem 1 (first-principles derivation of θ_n): The self-consistency condition $n\theta_n = \widetilde{Q}_n(\theta_n)$ is extended to all odd n and proved to have a unique solution (Proposition 1). The closed-form asymptotic $\theta_n \to 8\pi^2/[(\pi+4)^2 n^2]$ is derived analytically (Eq. (6)). The physical principle elevating the self-consistency condition to a theorem remains the primary open problem.

Problem 2 (cinquefoil mass spectrum): The five $\mathbb{Z}_5$ mode masses are $\{3.6, 17.6, 838.3, 1068, 2677\}$ MeV (Table 4). The two light modes are anti-phase near-zeros with no PDG counterpart; the three heavier modes bracket the ω–ϕ meson region with 5–9% deviations.

Problem 3 (meaning of the $3+2$ split): Theorem 1 proves that the number of negative-amplitude modes is $n_- = \lfloor 5n/8 \rfloor - \lceil 3n/8 \rceil + 1$, equal to 2 for $5 \le n \le 11$. This is a consequence of the CDFT coefficient $c_n = \sqrt{2}$ placing the amplitude zeros at the 45° diagonals, so the negativity arc spans exactly one quarter-circle.

Problem 4 (higher families): Complete mode spectra for $n = 7, 9, 11, 13$ are computed (Tables 6–9). The $n^{7/4}$ mass hierarchy (Paper VII) is verified for all families. Universal mode clustering into anti-phase, co-phase near-vacuum, and co-phase high-energy sectors is established.

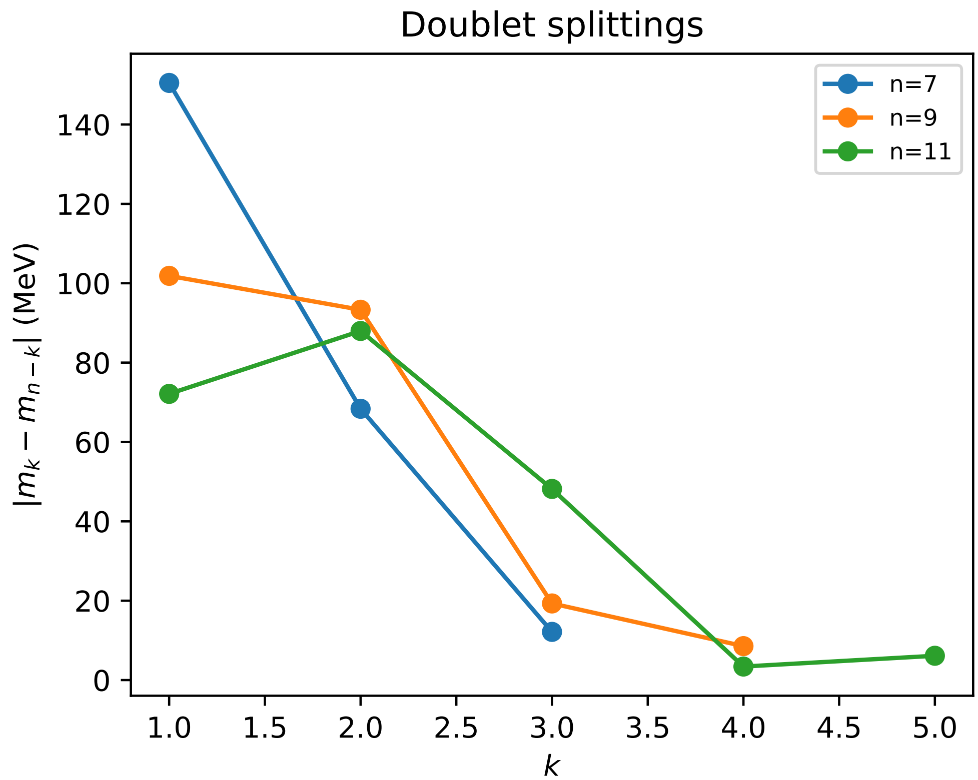

Figure 4: Doublet splittings $|m_k - m_{n-k}|$ across the hierarchy. The pairing grows progressively tighter as θ_n decreases.

Supplementary Material

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

Data and Code Availability

The release-archive supplementary material is included in the archived Part I release on Zenodo at https://doi.org/10.5281/zenodo.20250820. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at https://github.com/bampita-bico/CDFD-Part-I-Release. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

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Declarations

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing–original draft, and writing–review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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